Nguyen Manh Hiep

Software Engineer

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SUMMARY

I am seeking a Software Engineer position where I can apply my strong technical foundation, systems thinking, and business analysis skills to develop software products with practical value. I am interested in turning ideas and business problems into complete technology solutions that can be deployed in real-world business environments. With problem-solving ability, experience in building scalable systems, and an independent working mindset, I aim to contribute to stable, effective products that create clear impact for the organization.

PROJECTS

Notarex - Notary Document OCR & Digitization 2026 - 2026

Software Engineer (Full-stack + AI Orchestration + Infrastructure):

- **Description:** A platform that digitizes and manages notarial documents, allowing notary offices to extract data from scanned papers with AI OCR, edit the results online, and automatically generate finished Word documents.
- **Technologies & Tools:** Next.js 14, TypeScript, Ant Design, Zustand, TanStack Query, FastAPI (Python 3.11), SQLAlchemy 2.0 async, PostgreSQL 16, Celery, Redis, MinIO, OnlyOffice, docxtpl, Nginx, Docker, GitHub Actions.
- **Key Responsibilities & Contributions:**
  - **End-to-end Feature Development:** Built configurable form templates, an AI-powered OCR extraction flow, and an edit–save–download document workflow integrated with OnlyOffice secured by JWT, along with automatic DOCX generation using docxtpl.
  - **Authorization Design:** Implemented a 3-role permission model (admin / director / employee) scoped per notary office to keep data isolated and access clearly controlled.
  - **Background Processing:** Designed a Celery + Redis job queue using a ZSET pending queue and an atomic active-counter, with heartbeat reconciliation to detect and recover zombie jobs, ensuring OCR tasks run reliably at scale.
  - **Infrastructure & DevOps:** Managed file storage on MinIO (S3-compatible), containerized the system with Docker behind Nginx, and set up CI/CD pipelines with GitHub Actions.

SenAI - Real-time Vietnamese-English Meeting Interpreter 2026 - 2026

Software Engineer (Full-stack + Real-time AI + Infrastructure) :

- **Description:** A real-time bilingual meeting platform that provides per-speaker speech recognition, Vietnamese-English translation, live subtitles, translated chat, automatic meeting minutes, and deployment options for both on-premise and commercial AI providers.
- **Technologies & Tools:** Next.js 16, React 19, TypeScript, Tailwind v4, FastAPI, PostgreSQL 16, pgvector, LiveKit, WebSocket, vLLM, Docker, Nginx, GitHub Actions, systemd, Amazon SES.
- **Key Responsibilities & Contributions:**
  - **Frontend & Product UI**: Built the full Next.js web application with 7 feature areas including landing, authentication, live meeting room, rooms, notes, reports, and admin screens; implemented a custom design system, Vietnamese-English i18n, light/dark mode, and PWA support.
  - **Real-time Meeting System:** Developed the LiveKit-based meeting room with camera, microphone, screen sharing, translated two-way chat, waiting-room approval, host controls, QR invites, speaker mode, and runtime meeting settings updates.
  - **Translation Pipeline:** Implemented independent per-microphone ASR and machine-translation pipelines to keep speaker audio separated and stream translated subtitles during the meeting; supported fast translation plus LLM refinement with glossary context.
  - **Backend & Data Layer:** Built and maintained FastAPI services for meeting configuration, token generation, WebSocket authentication, model routing, transcript storage, and bilingual meeting minutes; migrated the data layer from SQLite to PostgreSQL 16 + pgvector while preserving existing public function signatures.
  - **AI/LLM Integration:** Integrated a pluggable model router for ASR, fast translation, refined translation, TTS, and LLM capabilities, allowing runtime provider switching between self-hosted models and commercial APIs with fallback handling.
  - **Deployment & Quality Assurance:** Set up CI/CD with GitHub Actions and pull-based VPS deployment behind Nginx and systemd; supported Docker-based services for LiveKit and PostgreSQL; authored and maintained 37 UAT suites with 147 real non-mocked tests.

EDUCATION

HUST - Hanoi University of Science and Technology 2021 - 2026

Computer Science

HONORS & AWARDS

Hackathon Third Prize at Vietnam AI Innovation Challenge 2026

TECHNICAL SKILLS

AI-assisted development	Claude Code (agents, skills, hooks), Cursor, ChatGPT, Gemini, GitHub Copilot, MCP servers, prompt & context engineering, spec-driven AI workflows, RAG, LLM orchestration, vLLM self-hosting
Frontend	React 19, Next.js 16/14, TypeScript, Tailwind v4, Ant Design, Zustand, TanStack Query, PWA / offline-first, i18n
Backend	FastAPI, Node.js, NestJS / ExpressJS, Spring Boot, Python, Java, REST, WebSocket, WebRTC (LiveKit), Celery, Redis, JWT /OAuth, RBAC
Databases	PostgreSQL 16 (+ pgvector), MySQL, MariaDB, SQLAlchemy 2.0 async, schema migrations, vector search & embeddings, MinIO (S3)
DevOps & Cloud	Docker & Docker Compose, Nginx, systemd, TLS, VPS deployment, GitHub Actions CI/CD, Cloudflare, Amazon SES
Engineering practice	Systems thinking, software architecture, SOLID, OOP, design patterns, real (non-mocked) integration & security testing, requirements analysis, SRS / Use Case / ERD
Soft Skills	Critical thinking, technical research, cross-functional teamwork, and quick adaptability to new technologies.